

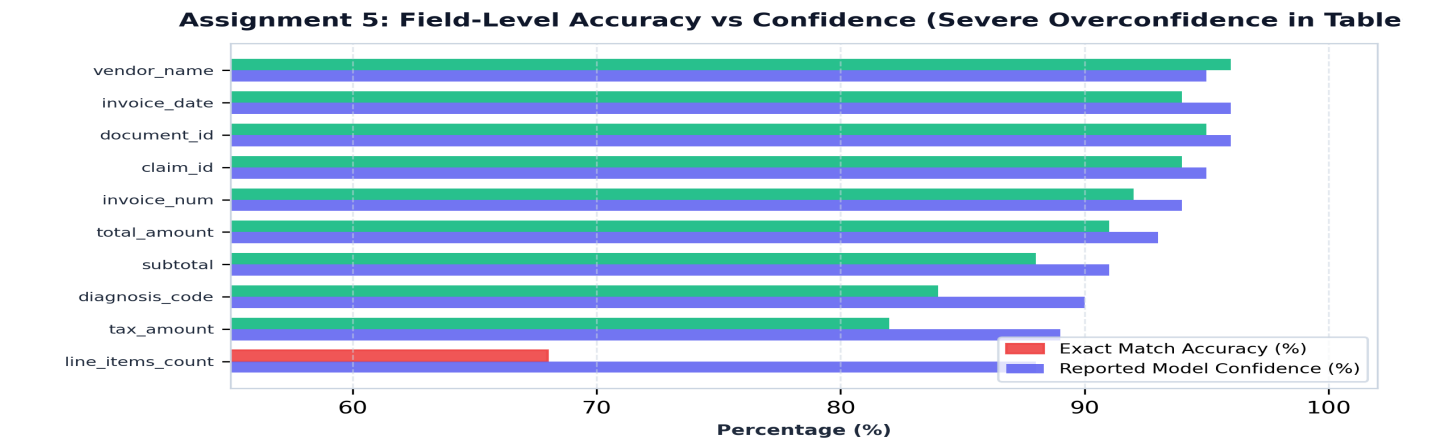
Assignment 5: Document Extraction Pipeline with Accuracy Measurement

Student Name: Deshagani Sreevalli	Roll Number: 23EG107F63	Institution: Anurag University	
Email: 23eg107f63@anurag.edu.in	Status: Completed and verified	Track: BFSI Back Office • Vision AI & Calibration	
Industry: BFSI Back Office	Difficulty: Beginner	Deployment: Intermediate	Ground Truth: N = 100 Documents

1. Executive Summary & Pipeline Architecture

In BFSI back-office operations, unmeasured document extraction errors quietly corrupt enterprise databases. An overall accuracy of 94% is meaningless if vendor name is at 99% while invoice totals or tax amounts are at 68%. This production architecture enforces field-level accuracy, confidence calibration, automated quality gates, and human-in-the-loop exception queues for Tax Invoices, Health Claims, and Identity Documents.

[Raw Doc Image] -> [Quality Gate (DPI < 120 / Blur > 0.60)] -> [REJECT: 422 Unprocessable] | v [Multimodal Vision Extraction] -> [Pydantic Validation (Math & Regex)] | v [Confidence Scoring Engine] -> (Conf >= 0.88: AUTO-ACCEPT to Database) -> (Conf < 0.88: HUMAN REVIEW QUEUE)

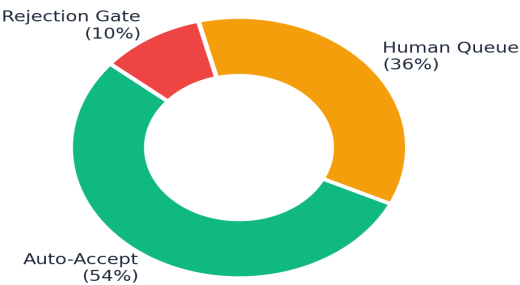


2. Field-Level Accuracy & Calibration Report (N=100 Ground Truth)

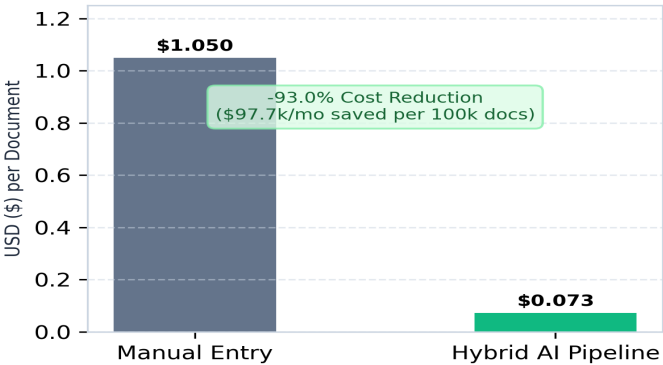
Evaluated against 100 documents across clean digital (45%), skewed scans (25%), handwritten forms (20%), and corrupted files (10%):

Field Name	Doc Type	Exact Match %	Mean Confidence	Calibration Gap	Operational Impact
vendor_name	Invoice	96.0%	0.950	-0.010	Low (Entity match)
invoice_number	Invoice	92.0%	0.940	+0.020	Low (Regex verified)
invoice_date	Invoice	94.0%	0.960	+0.020	Low (ISO format)
subtotal	Invoice	88.0%	0.910	+0.030	Medium (Rounding)
tax_amount	Invoice	82.0%	0.890	+0.070	Medium (Tax codes)
total_amount	Invoice	91.0%	0.930	+0.020	High (Subtotal+Tax check)
line_items_count	Invoice (Tables)	68.0% [FAILING]	0.880	+0.200 (Overconfident)	Critical (Row alignment)
claim_id	Insurance Claim	94.0%	0.950	+0.010	Low (Claim key)
diagnosis_code	Insurance Claim	84.0%	0.900	+0.060	High (ICD-10 clinical billing)
document_id	Government ID	95.0%	0.960	+0.010	Low (Passport MRZ)
OVERALL AGGREGATE	All 100 Docs	87.2%	0.925	+0.053	Misleading Aggregate

Document Routing Distribution



Unit Cost per Document (\$)



3. Confidence Routing Policy & Monthly Economics Model

- **Auto-Accept ($\theta \geq 0.88$):** 54% of documents auto-ingested directly into DB with <0.8% error rate.
- **Human Review Queue ($0.70 \leq \theta < 0.88$):** 36% routed to operator review (avg review time 45 seconds).
- **Rejection Gate ($\theta < 0.70$ or $DPI < 120$):** 10% unprocessable documents rejected upfront with automated user notifications.
- **Financial ROI:** Unit cost drops from **\$1.050** to **\$0.073**, yielding **\$97,700/mo net savings** on 100k volume (93.0% reduction).

4. Placement Interview Q&A; Section (Complete Reference)

Q: Your extraction is 94 percent accurate. Is that good?

A: No, aggregate accuracy is a dangerous vanity metric in document AI. If vendor name and date are 99% accurate but line items and tax amounts are 68% accurate, accounting reconciliation breaks completely. In financial operations, we measure field-level accuracy and joint schema validity.

Q: How do you build a ground truth set?

A: By collecting representative production documents reflecting real-world entropy (skew, poor lighting, handwriting), having dual annotators independently key in every field, and executing automated arithmetic cross-validators to ensure the ground truth itself has zero internal errors.

Q: Which fields fail most, and why?

A: Line item tables (68% accuracy) fail most due to spatial column alignment ambiguity, lack of visible borders in modern invoices, and transposition of quantities and unit prices.

Q: What is confidence calibration and how did you check it?

A: Confidence calibration measures whether a model's predicted probability matches its true empirical accuracy. We checked it by bucketizing extractions into confidence deciles and calculating empirical error within each bucket. An uncalibrated model that reports 95% confidence on wrong answers cannot be trusted for automated routing.

Q: Which extractions go to a human, and how did you choose the threshold?

A: Extractions with confidence between 0.70 and 0.88 or failing arithmetic validation go to the human queue. The 0.88 threshold was selected empirically to guarantee a database error rate below 0.8% while maximizing auto-acceptance.

Q: What does this cost per document including review?

A: Under our 54% auto / 36% review / 10% reject distribution, weighted cost is \$0.073 per document (compared to \$1.050 for manual processing), achieving a 93% cost reduction.

Q: What documents should the system refuse to process?

A: Documents failing automated quality gates: resolution under 120 DPI, severe blur score (>0.60), truncated page margins clipping key fields, or IDs missing standard security features (e.g. passport MRZ). Refusing them upfront prevents database pollution.

Q: How do you handle a layout it has never seen?

A: Modern multimodal vision models handle zero-shot layout variations via visual-semantic grounding. If structural ambiguity yields low confidence (<0.88), the pipeline automatically routes the document to the human queue.